

a form interpretable by the underlying mandatory security mechanisms of the operating system. Without such a framework, there can be no real confidence that the mandatory security mechanisms will provide the desired security properties. An operating system that provides mandatory security may nonetheless suffer from the presence of high-bandwidth covert channels. This is an issue whenever the mandatory security policy is concerned with confidentiality. This should not, however, be a reason to ignore mandatory security. Even with covert channels, an operating system with basic mandatory controls improves security by increasing the required sophistication of the adversary.

Once systems with basic mandatory controls become mainstream, covert channel exploitation will become more common and public awareness of the need to address covert channels in computing systems will increase.

In any system that supports mandatory security, some applications require special privileges in the mandatory policy in order to perform some security-relevant function. Such applications are frequently called trusted applications because they are trusted to correctly perform some security-related function and because they are trusted to not misuse privileges required in order to perform that function. If the mandatory security mechanisms of a secure operating system only support coarse-grained privileges, then the security of the overall system may devolve to the security of the trusted applications on the system. To reduce the dependency on trusted applications, the mandatory security mechanisms of an operating system should be designed to support the principle of least privilege.

Type enforcement is an example of a mandatory security mechanism that may be used both to limit trusted applications to the minimal set of privileges required for their function and to confine the damage caused by any misuse of these privileges.

The mandatory security mechanisms of an operating system may be used to support security-related functionality in applica-